Extended exercises 1 for Lecture "Theoretical Spectroscopy for systems implying periodic boundary conditions"

Exercise 1: Periodic Boundary Conditions, DFT, Geometry and Cell Optimization

based on exercises taken from the CP2K Homepage.

1.1: Generating an input structure for periodic boundary conditions

- Choose a system of interest and generate a start structure. For bulk systems, coordinate files can be downloaded from the Materials Project webpage. To download corresponding CIF files, you have to create an account, log in and click on the atom types of interest on the periodic table. To get a starting structure for CdSe, e.g., you would choose 'Cd' and 'Se'. After clicking on search, a list of possible crystal structures will be provided from which you can choose the desired one. Doing so, the unit cell of the chosen crystal structure is visualized and by clicking on the last item at the upper right corner of the visualization window, the corresponding CIF file can be downloaded to your local machine.
- Use scp to copy the CIF file to the Caucluster:
  `scp -r CdSe.cif suphcXXX@caucluster.rz.uni-kiel.de:~/`
- Log in to Caucluster:
  `ssh suphcXXX@caucluster.rz.uni-kiel.de`
- You can check that the CIF file has been copied to your $HOME directory with `ll` or `ls`.
- Switch to your $WORK directory and create a new directory for the computation, enter the directory and move the CIF file to this new directory
  ```
  cd $WORK
  mkdir Single_point_CdSe
  cd Single_point_CdSe
  mv ~/CdSe.cif .
  ```
  With `ls` or `ll` and `pwd`, you can again check that the last steps were successful.

1.2: Performing a single-point DFT computation implying periodic boundary conditions

- To perform a single-point energy DFT calculation, you can rely on the examplary input file `pd.inp` that is being provided in the directory
  `/zfshome/suphc356/TEACHING/TEACHING_1004D_XXXXXX/Inputs_Exercise1`
- Copy `pd.inp` to your current working directory
  `cp /zfshome/suphc356/TEACHING/TEACHING_1004D_XXXXXX/Inputs_Exercise1/pd.inp .`
- Modify `pd.inp` with vim: Open the file with vim,
  `vim pd.inp`
  switch to insert mode by typing "i", and go with the cursor to the "&SUBSYS" section, defining the structure of the system of interest. Replace the lattice parameters of the

unit cell as well as the Cartesian coordinates of the atoms in the unit cell by replacing "Pd.cif" with the corresponding CIF file of interest, e.g. "CdSe.cif".

- Furthermore, since we are using the Gaussian and plane waves (GPW) ansatz, a Gaussian basis and pseudo potentials have to be defined for each atom type requiring within the `&SUBSYS` section a `&KIND XX` section, in our case e.g.
  `&KIND Cd`
  `BASIS_SET DZVP-MOLOPT-SR-GTH`
  `POTENTIAL GTH-PBE-q12 &END KIND`
  Note that the acronym for the pseudopotential is atom-type specific, indicating the number of implied valence electrons with the ending "-qXX". To check the availability of the atom-type specific pseudopotentials, you can go to the Github directory of CP2K, https://github.com/cp2k/cp2k,
  and check in the subdirectory `data` the corresponding pseudopotential file `GTH_POTENTIALS`, listing all pseudopotentials e.g. in our case of CdSe:
  `Cd GTH-PBE-q12 GTH-PBE`
  `Se GTH-PBE-q6 GTH-PBE`

- Add `&KIND` sections for all atom types that your system of interest comprises to your input file, in our case of CdSe e.g.
  ```
  &KIND Cd
  BASIS_SET DZVP-MOLOPT-SR-GTH
  POTENTIAL GTH-PBE-q12 &END KIND
  &KIND Se
  BASIS_SET DZVP-MOLOPT-SR-GTH
  POTENTIAL GTH-PBE-q6 &END KIND
  ```

- Store your changes by leaving the insert mode of vim with escape and typing ":wq".

- You can check again your directory path with `pwd` and you check again that the input file has been modified as described by opening and closing it again with vim.

- To run a computation, you furthermore have to copy the bash script for slurm to your local directory,
  ```
  cp /zfshome/suphc356/TEACHING/TEACHING_1004D_XXXXXX/Inputs_Exercise1/run_with_caucp2k.sh .
  ```

- Open `run_with_caucp2k.sh` with vim and modify the line that specifies input and output files given to CP2K according to your needs,
  `mpirun -np 32 cp2k.psmp -i XXX.inp -o cp2k.out`

- Submit the job with
  `sbatch run_with_caucp2k.sh`

- Slurm will produce two output files `test.err` and `test.out` containing information on the ressources that were required by the computation (memory, computation time) and, optionally, on errors that could have occured when running the computation. Furthermore, the output of the computation will be written to `cp2k.out`. Open `cp2k.out` and check that CP2K has printed no warnings at the end of the output file,
  "`The number of warnings for this run is :   1`".

- Check that the self-consistent field iterations have converged, with CP2K printing in this case the confirmation
  `*** SCF run converged in XX steps ***`

- Check the total energy of the system by searching for "`Total energy:`".

1.3: Geometry and cell optimization

- To perform a geometry optimization, the keyword `RUN_TYPE` in the input section `GLOBAL` has to be modified from `ENERGY` to `GEO_OPT`. Furthermore, a section called `MOTION` can be added to define e.g. four convergence thresholds via the keywords `MAX_DR`, `MAX_FORCE`, `RMS_DR`, and `RMS_FORCE`.

- An examplary input file for a geometry optimization is provided by the file `pd_geoopt.inp` in the directory
  `/zfshome/suphc356/TEACHING/TEACHING_1004D_XXXXXX/Inputs_Exercise1`.

- Create a new directory in your $WORK directory
  `mkdir Cd_Se_geoopt`

- Switch to the new directory and check that this was successfully done with `pwd`
  `cd Cd_Se_geoopt`

- Copy `pd_geoopt.inp` to your new directory,
  `cp /zfshome/suphc356/TEACHING/TEACHING_1004D_XXXXXX/Inputs_Exercise1/pd_geoopt.inp .`

- Copy the input and output files of Exercise 1.2 to your new directory,
  `cp -r $WORK/Single_point_CdSe/* .`

- Compare the old input file and the new input file using `vimdiff`
  `vimdiff pd_geoopt.inp pd.inp`

- Modify again the "&SUBSYS" section adjusting to your system of interest.

- Your new input file should now define
  `RUN_TYPE GEO_OPT`
  as well as a section to define the thresholds for the to-be-performed geometry optimization,
  ```
  &MOTION
  &GEO_OPT
  OPTIMIZER CG
  MAX_ITER 200
  MAX_DR 1.0E-03
  MAX_FORCE 1.0E-03
  RMS_DR 1.0E-03
  RMS_FORCE 1.0E-03
  &END GEO_OPT
  &END MOTION
  ```

- Submit the computation by adjusting `run_with_caucp2k.sh` with vim, modifying the line that specifies input and output files given to CP2K according to your needs,
  `mpirun -np 32 cp2k.psmp -i XXX.inp -o cp2k.out`
  and by submitting the job with
  `sbatch run_with_caucp2k.sh`

- When changing the keyword `GEO_OPT` to `CELL_OPT` as done in `pd_cellopt.inp`, and by adding the keyword

```
STRESS_TENSOR ANALYTICAL
```

to the `&FORCE_EVAL` section, the unit cell is optimized as well. Run a computation to optimize the unit cell and compare the computation with the results obtained from performing a geometry optimization, implying a unrelaxed unit cell.

1.4: Vibrational analysis

- The computation of vibrational frequencies is requested by changing the run type from geometry optimization (`RUN_TYPE GEO_OPT`) to vibrational analysis (`RUN_TYPE VIBRATIONAL_ANALYSIS`) in the CP2K input file as well as by adding a section *&VIBRATIONAL_ANALYSIS* and a print comment (`&PRINT/&MOMENTS ON`) to switch on the computation of corresponding dipole moments.

- To compute vibrational frequencies, create a new directory,
  ```
  mkdir VIB_ANALYSIS
  ```
  copy all previously generated results to the new directory,
  ```
  cp $WORK/Cd_Se_geoopt/* .
  ```
  and modify the input file, adjusting the run type as well as appending the following section after the `&FORCE_EVAL` section,
  ```
  &VIBRATIONAL_ANALYSIS
  INTENSITIES T
  FULLY_PERIODIC TRUE
  DX 0.001
  &END VIBRATIONAL_ANALYSIS
  ```
  Furthermore, switching on the printing of dipole moments implying periodic boundary conditions is achieved by adding the following section within the `&FORCE_EVAL` section:
  ```
  &PRINT
  &MOMENTS ON
  PERIODIC .TRUE.
  &END MOMENTS
  &END PRINT
  ```

- After checking that the input file has been successfully changed as described, submit the computation with slurm.

- You can compare and visualize the obtained vibrational frequencies and corresponding intensities using the Fortran code `newspecmatch`.

- To use newspecmatch, insert the following two lines in your `.bashrc` file and source the file:
  ```
  export newspecmatchhome=/zfshome/suphc356/newspecmatch/src
  export PATH=$PATH:$newspecmatchhome
  ```
  Open the `.bashrc` file with `vim` and append the two lines.

- Newspecmatch enables with the option `-plot` to plot the spectral function:
  ```
  newspecmatch spectra_1.dat -plot -fscal 0.982 -range 0 3500.
  ```
  Furthermore, vibrational spectra can also be compared with the norms closer to 1 indicating a good match:
  ```
  newspecmatch spectra_1.dat spectra_2.dat
  ```

- You can scale the obtained frequencies by adjusting the global scaling factor `fscal` and plot the spectra again using the plot keyword and `gnuplot spectra.gnup`.

1.5: Absorption spectra

- To compute an absorption spectrum, you have to modify the CP2K input file according to the examplary input file `stda_absorption.inp` given in directory `/zfshome/suphc356/TEACHING/TEACHING_1004D_XXXXXX/Inputs_Exercise2.`, implying that the `RUN_TYPE` is switched back to `ENERGY` and that a `&PROPERTIES` section is added,

```
&PROPERTIES
&TDDFPT
KERNEL sTDA
&sTDA
FRACTION 0.0
DO_EXCHANGE F
DO_EWALD T
&END sTDA
NSTATES 10
MAX_ITER 100
CONVERGENCE [eV] 1.0e-7
RKS_TRIPLETS F
&END TDDFPT
&END PROPERTIES
```

- Choosing `sTDA` as option for the keyword `FULL_KERNEL` implies a semi-empirical TDDFT kernel, not computing the exact 2-electron-4-center integrals for Coulomb and exchange interactions, but relying on a semi-empirical repulsion operator, drastically approximating the Coulomb and exchange integrals and thus saving computation time. If choosing this kernel, which is recommended due to the limited time in class, the subsection `&sTDA` has to be added, specifying Ewald summation when implying periodic boundary conditions and choosing the amount of exact exchange with the keyword `FRACTION`. The latter should be chosen depending on the underlying ground-state DFT reference. In case of an underlying PBE functional, as in our case, exchange contributions have to be switched off, explicitly adding the keyword `DO_EXCHANGE`.

- Create a new directory, copy all old input and output files to the new directory, modify the CP2K input file as described above and submit the computation with sbatch.

- You can visualize the absorption spectrum using the Jupyter Notebook `Visualizing_Absorption_Spectrum.ipynb`. The Jupyter Notebook requires as input the excitation energies and oscillator strengths that were computed with CP2K, copied to a file called `spectra.inp`, listing in a first column excitation energies and in a second column the corresponding oscillator strengths, e.g.

```
4.56181& 0.03652230 &
4.92689& 0.0008994580 &
5.21246& 0.09602060 &
5.89121& 0.02028950 &
6.11704& 0.209590&
6.51699& 0.001807480 &
6.5964& 0.009386190 &
6.63204& 0.1006890 &
6.69911& 0.06938320 &
6.89093& 0.08802080 &
```